Connect the forces to the final position

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Constant net force / ES: fuerza neta constante / PT: força resultante constante

Gives constant acceleration: a straight v–t line.

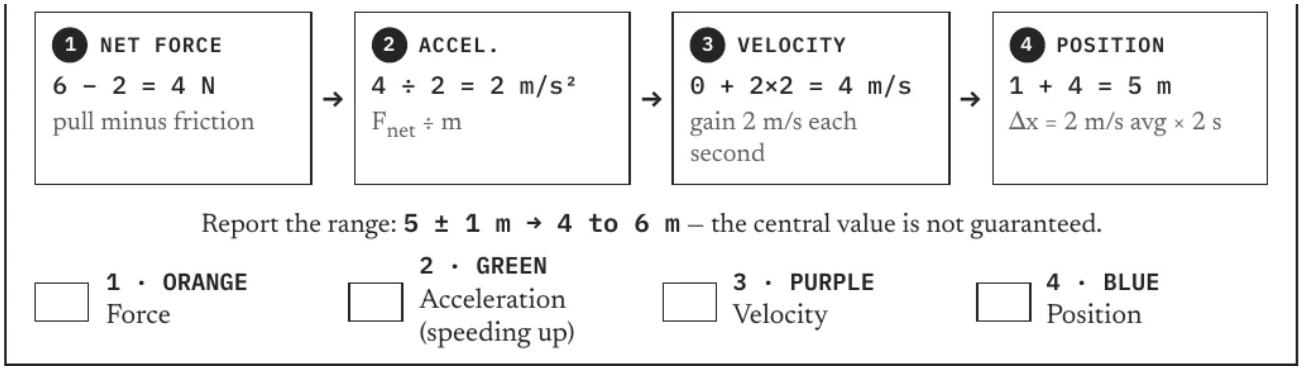
Also on your note page: Force and motion, Zero net force, Range.

B READ THE RELATIONSHIP

Net force determines acceleration. To find position, you also need the starting motion.

Force → acceleration → velocity → position

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement.

Given F net = 0 N v = +2 m/s t = 3 s

Rule a = 0 → v constant $\Delta x = v \times t$

Work v stays +2 m/s $\Delta x = 2 \times 3$

Answer +2 m/s, +6 m

E SAVE FOR THE FINAL MEETING

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

Read the physics: Constant net force

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Divide net force by mass to find acceleration. For constant acceleration, use the initial velocity and elapsed time to find final velocity.

02 HOW IT WORKS

Use the signed area under the velocity–time graph for displacement. Add displacement to starting position. A section below zero subtracts from displacement but still counts toward distance.

03 WHAT TO CHECK

With zero net force a moving object continues at its initial velocity. With constant nonzero net force its velocity changes by the same amount each second.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart has zero net force and already moves at +2 m/s. After 3 s, find its velocity and displacement. +2 m/s, +6 m.

05 CONNECT TO XJ

The chain is complete: net force → acceleration → velocity → position. That is the whole route from the data on XJ to a forecast of where it will arrive.

SAY IT BEFORE YOU WRITE

What do you need besides force and mass to predict position? From force I can find ___. To find position I also need ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Constant net force / ES: fuerza neta constante / PT: força resultante constante

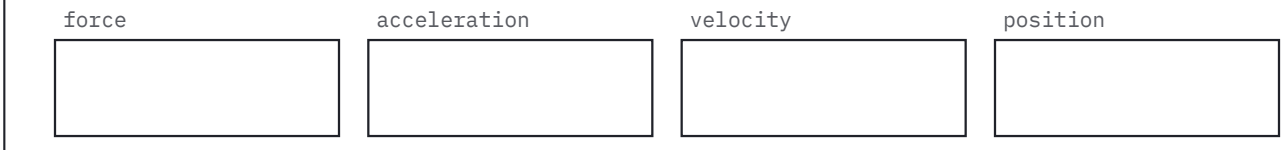
Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, constant net force means …	A cart's mass stays the same. If its net force stays the same in size and direction, what else stays the same? The cart's ___ stays the same.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

Labels you can use: force · velocity · position.



Tell a partner: From force I can find ___. To find position I also need ___.

On assignment: Constant net force

SECTION E / BACK / FINAL MEETING ONLY Name _____

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Mission systems engineer**

Mark which part of the forecast you can release now. Send the operator a request for the missing measurements.

☐ **News assignment editor**

Prepare a "known / still needed" graphic. Tell the anchor which result can be reported now and which must wait.

Word bank: force · velocity · position.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."